

DSA Day 5: Structures (struct -> class)

Book: Data Structures Through C in Depth - Ch2 §2.4 · pp 34-47 | Code in Python

Overview

A C struct groups fields; in Python use a class or dataclass. A list of objects mirrors an array of structs. Self-referential nodes are the basis of linked structures.

Key concepts to master

- Fields -> attributes
- Array of struct -> list of objects
- Nested structures
- Self-referential Node preview

Python implementation

```
class Student:
    def __init__(self, name, roll, marks):
        self.name=name; self.roll=roll; self.marks=marks
s=[Student('A',1,80),Student('B',2,91)]
top=max(s, key=lambda x:x.marks)
```

Complexity

Field access $O(1)$.

Common pitfalls

- Mutable default arguments in Python
- Forgetting self

Practice

- Find topper among 5 students
- Why do nodes reference their own type?

